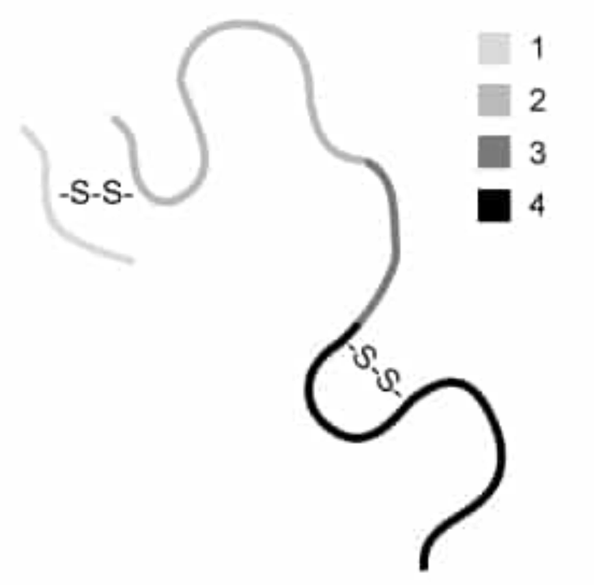
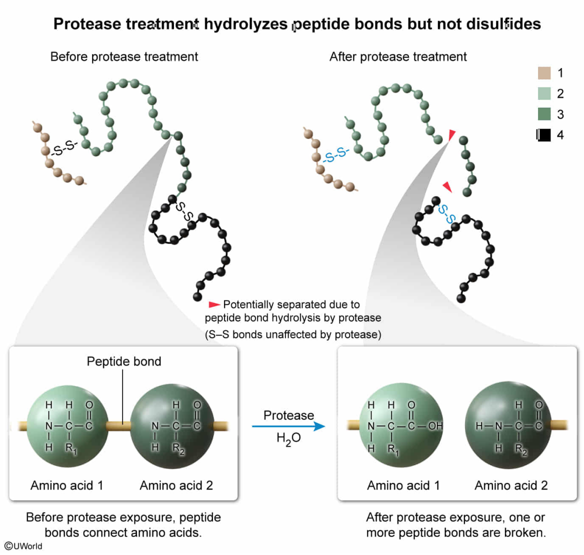
A certain multi-subunit protein consists of four regions, some of which contain cysteines that form disulfide bonds, as depicted in the diagram below. Treatment with a protease can cleave the protein into smaller fragments. Given this method, which regions of the protein shown below are NOT likely to be separated by protease treatment?



- A. Regions 1 and 2 [52%]
- B. Regions 1 and 3 [6%]
- C. Regions 2 and 3 [34%]
- D. Regions 3 and 4 [6%]

Correct
51% answered correctly

Explanation:



Proteins consist of amino acids linked together by peptide bonds, which join the carboxyl group of one amino acid to the amino group of the next amino acid in a polypeptide. **Proteases** are enzymes that **facilitate** the **breaking of peptide bonds** in a hydrolysis reaction.

The question describes the structure of a particular multi-subunit protein and notes that certain regions contain disulfide bonds. The question asks for regions of the protein that are **not** likely to be separated from each other by treatment with a protease. Because proteases specifically target peptide bonds, a **protease will not separate** regions **1 and 2**, which are only **linked by** a **disulfide bond**.

(Choice B) Although regions 1 and 2 are only linked by a disulfide bond, regions 2 and 3 are part of the same subunit of the protein and are composed of amino acids linked by peptide bonds. A protease could break one or more of the peptide bonds in region 2, thereby separating region 1 from region 3.

(Choices C and D) Regions 2, 3, and 4 are all part of a subunit of the protein and are composed of amino acids linked by peptide bonds. Therefore, a protease could break one or more of the peptide bonds linking amino acids in these regions and separate one region from another.

Educational objective:
Proteases are enzymes that break peptide bonds in proteins via a hydrolysis reaction. During peptide hydrolysis, a water molecule is used to cleave the C–N bond in the peptide linkage.

Time spent:0 sec	**QID:**403090
Subject: Biochemistry	**System:** Enzymes